

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EC) Examination

April - May 2015

EC 204 / EC204.01 Electronics Instruments & Measurement

Date: 29.04.2015, Wednesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Define Followings. [07]

- | | |
|------------------|-------------------------|
| (i) Accuracy | (ii) Limiting Error |
| (iii) Resolution | (iv) Precision |
| (v) Sensitivity | (vi) Significant Figure |
| (vii) Error | |

Q - 2 (a) A set of independent current measurements was taken by six observations and recorded as 12.8mA, 12.2mA, 12.5mA, 13.1mA, 12.9mA and 12.4mA. [04]

Calculate (a) Arithmetic mean (b) Deviations from the mean.

(b) Explain the basic circuit for measurement of frequency showing Gate Control F/F. [05]

(c) Draw the structure of an LED and explain its operation. [05]

OR

Q - 2 (a) A $4\frac{1}{2}$ digit voltmeter is used for measuring voltage measurements, [04]

- (i) Find its Resolution
- (ii) How would 12.98V be displayed on 10v range?
- (iii) How would 0.6973 be displayed on 1V and 10V ranges.

(b) Explain the operation of period measurement with the multiple average mode. [05]

(c) What are the techniques used to reduce the input frequency (High frequency)? [05]

Q - 3 (a) Explain the working principle of a Q meter. Also outline the factors that cause errors during a Q measurement. [05]

(b) What is DAS? Explain with help of a block diagram. [05]

(c) State three major categories of errors. [04]

OR

- Q - 3 (a)** How can a Q meter be used for the measurement of stray capacitance and impedance of a circuit? [05]
- (b)** Explain the methods of signal conditioning. [05]
- (c)** Define the sensitivity and resolution of a digital meter. [04]

SECTION – II

- Q - 4 (a)** Define transducer. What are the differences between active and passive transducer? [04]
- (b)** What are limitations of Wheatstone bridge? [03]
- Q - 5 (a)** A Maxwell bridge is used to measure inductive impedance. The bridge constants at balance are :

$$C_1=0.01\mu F, R_1= 470k\Omega, R_2 = 5.1k\Omega, R_3= 100k\Omega$$

Find the series equivalent of the unknown impedance.

- (b)** Draw the basic block diagram of an oscilloscope and explain the function of each block. [05]
- (c)** Explain Kelvin bridge in brief. [04]

OR

- Q - 5 (a)** The impedances of the basic ac bridge are given as follows: [05]

$$Z_1 = 100\Omega \angle 80^\circ \text{ (inductive impedance)} \quad Z_3 = 400\Omega \angle 30^\circ \text{ (inductive impedance)}$$

$$Z_2 = 250\Omega \text{ (pure resistance)} \quad Z_4 = \text{unknown}$$

Determine the constants of the unknown arm.

- (b)** Write a short note on digital storage oscilloscope (DSO). [05]
- (c)** With neat diagram explain Wien bridge and derive frequency measurement equation. [04]
- Q - 6 (a)** Write a short note on [07]
- (i)** Load Cell
- (ii)** Thermistors
- (b)** Describe construction and operation of Linear Variable differential Transducer (LVDT) with necessary figures. [07]

OR

- Q - 6 (a)** Describe the operation of a Piezo-electric transducer. [07]
- (b)** Derive an expression for the gauge factor in case of bonded wire strain gauge. [07]
